LEVEL UP ACADEMY

Class 10 | CBSE / GSEB | Science

ANSWER KEY & MARKING SCHEME

Daily Practice: Ch 8 Heredity

Chapter / Topic: Heredity · Week 4 of the 90-day plan

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DAY 1: HEREDITY AND MENDEL'S BASICS

Q1. Which of these is NOT one of the pairs of contrasting traits Mendel studied in garden peas?

Answer: (d) red and black seeds. (Seed colour in his work was yellow or green.)

Q2. When two pure-breeding plants with contrasting traits are crossed, the trait seen in all the F_1 plants is called:

Answer: (b) dominant

Q3. The genotype of a pure-breeding short pea plant is:

Answer: (c) tt: a recessive trait shows only when both copies are recessive

Q4. In a cross between Tt and tt pea plants, the percentage of short plants expected is:

Answer: (c) 50%: Tt × tt gives Tt and tt in a 1 : 1 ratio

Q5. What is heredity? What is a gene?

Answer: Heredity is the passing of traits (characters) from parents to their offspring. A gene is a section of DNA that carries the information for making one protein, and so controls a trait.

Q6. Every person has two copies of each gene. Where does each copy come from?

Answer: One copy comes from the father and one from the mother, through their germ cells (sperm and egg).

Q7. A pea plant with yellow seeds (Yy) is self-pollinated. Show the cross with a checkerboard, and give the genotypic and phenotypic ratios of the…

Answer: Parents: Yy × Yy. Gametes: Y, y from each parent.
Checkerboard: YY, Yy, Yy, yy.
Genotypic ratio: 1 YY : 2 Yy : 1 yy.
Phenotypic ratio: 3 yellow : 1 green.

Q8. Explain, with an example of each: (a) a pure-breeding plant, (b) a hybrid, (c) the F_2 generation.

Answer: (a) A plant that gives offspring with the same trait generation after generation when self-pollinated, e.g. a TT tall plant.
(b) The offspring of two pure-breeding parents with contrasting traits, e.g. Tt from TT × tt.
(c) The generation obtained by self-pollinating the F_1 plants, e.g. Tt × Tt gives F_2 in the ratio 3 tall : 1 short.

DAY 2: WORKING OUT CROSSES

Q9. Which cross will give offspring that ALL show the dominant trait?

Answer: (c) TT × tt: every offspring is Tt

Q10. The 9 : 3 : 3 : 1 ratio in the F_2 of Mendel's dihybrid cross shows that:

Answer: (d) the two traits are inherited independently

Q11. Plant height depends on a growth hormone. The gene for height controls:

Answer: (a) If the gene makes an efficient enzyme, more hormone is made and the plant grows tall.

Q12. A cross between two tall pea plants gives offspring in the ratio 3 tall : 1 short. What are the genotypes of the parents?

Answer: Both are Tt. Tt × Tt gives TT, Tt, Tt and tt: 3 tall to 1 short.

Q13. A cross between two pea plants gives 50% round seeds and 50% wrinkled seeds. What are the genotypes of the parents?

Answer: Rr × rr. The gametes are R and r from one parent, and only r from the other, so the offspring are Rr and rr in a 1 : 1 ratio.

Q14. Mendel's F_1 plants from the dihybrid cross had the genotype RrYy. (a) Write the four kinds of gametes they make. (b) What fraction of the F_2 seeds…

Answer: (a) RY, Ry, rY and ry
(b) 3/16
(c) 3/16
(The full F_2 ratio is 9 round yellow : 3 round green : 3 wrinkled yellow : 1 wrinkled green.)

Q15. A tall pea plant could be TT or Tt. How can you find out which it is? Explain with crosses.

Answer: Cross it with a short plant (tt).
If it is TT: TT × tt gives all Tt, so all the offspring are tall.
If it is Tt: Tt × tt gives Tt and tt, so about half the offspring are short.

DAY 3: SEX DETERMINATION, MIXED PRACTICE AND CHALLENGE

Q16. **Assertion (A):** The sex of a child is decided at the time of fertilisation. **Reason (R):** Whether the fertilising sperm carries an X or a Y chromosome…

Answer: (a) Both are true, and R explains A.

Q17. In some reptiles, the sex of the young is decided by:

Answer: (a) the temperature at which the fertilised eggs are incubated

Q18. Which genotype belongs to a pure-breeding plant with round, yellow seeds?

Answer: (b) RRYY: pure-breeding means both copies of each gene are the same

Q19. A couple has three daughters and is expecting a fourth child. Some relatives blame the mother for not having a son.

(i) An X chromosome: all her eggs carry X.
(ii) No. The father's sperm carries either X or Y. A sperm with X gives a girl (XX) and one with Y gives a boy (XY), so the sex is decided by the sperm, not the egg.
(iii) 50%, or 1 in 2. Each pregnancy is independent, so the three daughters born earlier do not change the chance.

Q20. A plant with round, green seeds (RRyy) is crossed with a plant with wrinkled, yellow seeds (rrYY). (a) Write the gametes of each parent and the…

Answer: (a) Gametes: Ry from the RRyy parent and rY from the rrYY parent. F_1: RrYy, round and yellow.
(b) 9/16
(c) Round yellow and wrinkled green. (This shows that the two traits are inherited independently.)

Q21. A trait can 'skip a generation': a grandparent and a grandchild show it, but the parent in between does not. Explain how this can happen.

Answer: The trait is recessive. The parent got one recessive copy from the grandparent but also a dominant copy, so the trait stayed hidden. If the child then receives a recessive copy from both parents, it shows the trait again.